Queries, logical operators and big data

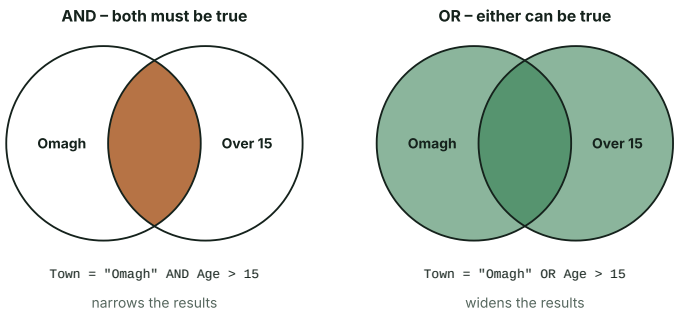
L10 FACT FILE

Specification: students should be able to

- Extract data from a database structure using simple query structures and using the following logical operators: <, >, =, <=, >=, AND, OR and BETWEEN
- Predict the records returned by a query with given criteria
- Demonstrate understanding of big data, referring to volume, velocity and variety
- Describe the difference between structured and unstructured data
- Demonstrate understanding of the need for data analytics to interpret big data

CCEA fact file: Fact File 08: Database Applications (2)

Key words: query, criteria, query design grid, comparison operator, logical operator, Boolean, AND, OR, BETWEEN, <>, big data, volume, velocity, variety, structured data, unstructured data, data analytics, data mining



`Town = "Omagh" AND Age > 15`

narrows the results

`Town = "Omagh" OR Age > 15`

widens the results

BETWEEN 10 AND 20 includes both 10 and 20.

Venn diagrams: AND returns records in both sets, OR returns records in either

Queries and comparison operators

A query is a question asked of a database. It checks every record and returns only those that meet the criteria, which are the conditions typed into the query. Most criteria use comparison operators. The equals sign finds an exact match, such as Club = "Lisnaskea". Less than (<) and greater than (>) exclude the value itself, so Goals > 8 does not return a player with exactly 8 goals. Less than or equal to (<=) and greater than or equal to (>=) include the value, so Score >= 50 returns a score of 50. Not equal to (<>) returns everything except the value given. Each comparison gives a Boolean result: the record either meets the condition (true) or does not (false). When writing criteria, put text values in quotation marks, and type them exactly as they are stored in the table. In Access, dates are written between hash signs, such as #01/03/2026#, so that they are treated as dates rather than text.

AND, OR and BETWEEN

Logical operators join two or more conditions together. With AND, both conditions must be true for the record to be returned, so AND usually reduces the number of results. For example, Club = "Lisnaskea" AND Goals > 10 only finds Lisnaskea players who have also scored more than ten goals. With OR, a record is returned if at least one condition is true, so OR usually increases the number of results. Club = "Lisnaskea" OR Club = "Irvinestown" finds players from either club. BETWEEN finds values within a range, and the range includes both end values. Age BETWEEN 13 AND 14 returns players aged 13 and 14, and gives the same results as Age >= 13 AND Age <= 14. A common mistake is to use AND when OR is needed: Club = "Lisnaskea" AND Club = "Irvinestown" returns nothing, because one player cannot belong to both clubs at the same time.

Building a query in design view

In Access, queries are usually built in the query design grid. First add the table or linked tables you need. Then place each field you want in the Field row. The Table row shows which table the field comes from. The Sort row puts the results in ascending or descending order, and the Show tick box decides whether a field appears in the results. A field can be used for criteria but hidden, for example to search by club without showing the club name. Criteria are typed in the Criteria row under the field they apply to. Conditions typed on the same row are joined with AND. Conditions typed on different rows, using the Criteria row and the Or row, are joined with OR. Once the query is run, its results can be used as the source of a report. This is how a club could print a list of players for one team, sorted by surname, without changing the data in the table.

Big data and the three Vs

Big data means data sets that are so large, arrive so fast and are so mixed that ordinary database software cannot store or process them in a sensible time. It is usually described using three Vs. Volume is the huge amount of data, often measured in terabytes or petabytes, such as every loyalty-card purchase made in UK supermarkets. Velocity is the speed at which new data is created and must be processed, often in real time. A bank must check a card payment for fraud within a second, and a social media post should appear straight away. Variety is the many different types and formats of data, such as text, photos, video, GPS locations and readings from sensors. Some of this is structured data, organised in tables with fixed fields such as names and dates. Much of it is unstructured data, such as posts, emails and video, which has no fixed format and cannot easily be searched with normal database tools.

Why data analytics is needed

Collecting big data is only useful if something can be learned from it. Data analytics means using algorithms, programs or artificial intelligence to examine data and find patterns, trends and relationships. Data mining is a part of this: searching very large data sets for patterns that nobody knew were there. Analytics is needed because big data is far too large and changes too quickly for people to examine by hand, and because it is in many formats that ordinary queries cannot handle. It needs specialist software, powerful computers, often hired from cloud providers, and skilled analysts. Examples include a supermarket using sales data to decide which offers to send each customer, a streaming service recommending films, a bank spotting unusual spending that may be fraud, and a health trust predicting busy periods in emergency departments so that more staff are on duty. Analytics helps organisations make decisions based on evidence rather than guesswork.

T Key terms from the mark schemes

Query
A way of searching a database to extract the records that meet set criteria. It checks each record and lists the data that satisfies the criteria.
Credited: U1 2021 Q2, 2023 Q3

Criteria
The conditions a record must meet to be included in the results of a query, e.g. Age >= 18.

Comparison operator
A symbol used to compare a value with other data, giving true or false: < less than, <= less than or equal to, > greater than, >= greater than or equal to, = equal to, <> not equal to.

Logical operator
An operator such as AND, OR or BETWEEN that combines or tests conditions and gives a true or false (Boolean) result.

BETWEEN
A logical operator that selects values within a range, including both end values.

Group by
Organises query or report results by the values of a field, e.g. grouping members by squad.
Credited: U1 2021 Q2

Big data
Very large sets of data that could be mined for information but cannot be analysed by normal means because of their volume, velocity and variety.
Credited: U1 2021 Q8, 2022 Q1

Volume
The very large amount of data stored by organisations, e.g. terabytes of social media posts each day.

Velocity
The speed at which data is created and must be processed, often in real time.

Variety
The many different types and formats of data, such as text, images, video and sensor readings, which may be structured or unstructured.

Structured data
Data organised in a fixed format, usually tables of fields and records, so it is easy to search and analyse, e.g. dates, names, addresses, card numbers.

Unstructured data
Data with no fixed format that cannot easily be processed with normal database tools, e.g. social media posts, emails, videos and essays.

Data analytics
Using algorithms, programs or artificial intelligence to examine data and find patterns, trends and relationships so that conclusions can be drawn.

Data mining
Searching very large data sets with software to discover patterns and relationships that were not previously known.

Exam focus

Query questions usually give you a table and ask you to complete criteria in a query design grid, or to say which records a query returns: 2019 Q3(d), 2021 Q2(f) and 2024 Q3(e). Copy field names and values exactly as they appear in the table, put text in quotation marks, and check boundary values carefully: > 15 does not include 15. Remember that criteria on the same row mean AND, while criteria on separate rows mean OR. Big data has appeared as a three-mark question on volume, velocity and variety (2021 Q8(b)) and in the multiple-choice question in 2022 Q1. For each V, give a short definition and an example: one mark is often given for each V explained. Do not mix them up. Velocity is about speed, not size, and variety is about different types of data, not different amounts. If asked why analytics is needed, explain that big data is too large and complex to analyse by normal means and that its value comes from the patterns found.

Unit 1 2024 Q3	[11]	Field names and data types, database terms, key field, query design
Unit 1 2022 Q1	[9]	MCQ: storage, bit rate, real-time, compression, buffering, ADC, big data, backup
Unit 1 2021 Q8(b)	[3]	Big data: volume, velocity, variety
Unit 1 2021 Q2(d)–(f)	[5]	Export to csv, queries, grouping
Unit 1 2019 Q3(d)–(f)	[5]	Query criteria, reports, sorting, relationship type

? Check yourself

1. Write the criteria to find all members who are aged 18 or over. [1]
2. Write the criteria to find all bookings made between 1 March 2026 and 31 March 2026. [2]
3. Explain the difference between the AND and OR operators in a query. [2]

Answers: 1. >=18 (in the Age field) (1). Accept Age >= 18 or >17. 2. Between #01/03/2026# And #31/03/2026# (1 for BETWEEN or >= and <= used, 1 for both correct dates). Accept >=#01/03/2026# And <=#31/03/2026#. 3. AND: both/all conditions must be true for a record to be returned (1). OR: only one/at least one condition needs to be true (1).